AP CALCULUS AB

Unit 1 • Lesson 1-7 • Guided notes

Lesson 1-7 • Instantaneous Rates of Change

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Define the derivative at a point as the limit of a difference quotient.
- Compute derivatives of simple functions from the definition and estimate them from tables and graphs.
- Write the equation of a tangent line, and recognize where a function is not differentiable.

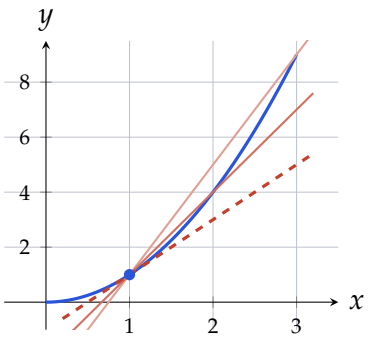
Vocabulary. instantaneous rate of change, derivative, $f'(a)$, tangent line, differentiable, corner, cusp, vertical tangent.

1. From average to instantaneous

Definition of the derivative

$$f'(a) = \lim_{h \to 0} _____ \quad \text{or} \quad f'(a) = \lim_{x \to a} _____ .$$

$f'(a)$ is the _____ of f at a and the _____ at $(a, f(a))$.



Secant lines from $(1,1)$ on $y = x^2$ approach the tangent line of slope 2.

Example 1 • From the definition

Find $f'(x)$ for $f(x) = x^2$, $g(x) = x^3$ and $k(x) = \frac{1}{x}$.

Example 2 • Tangent line

Write the equation of the tangent line to $y = x^3$ at $x = 2$.

Example 3 • Estimating from a table

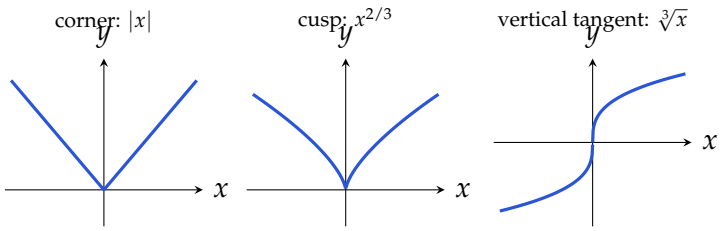
t (s)	0	2	4	6	8
$s(t)$ (m)	0	9	20	35	52

Estimate $s'(4)$ and interpret it.

2. Differentiability

Key idea

If f is differentiable at a, then f is _____ at a. The converse is _____.
f is **not** differentiable where its graph has a _____, a _____, a _____, or a _____.



Example 4 • A piecewise function

Is $f(x) = \begin{cases} x^2, & x \le 1 \\ 2x - 1, & x > 1 \end{cases}$ differentiable at $x = 1$?

Check your understanding

1. $\lim_{h\to 0} \frac{(3+h)^2-9}{h}$ is $f'(3)$ for $f(x) =$ _____, and equals ____.
2. $f(x) = |x - 2|$ is continuous at 2 but _____ there.